CDI COLLEGE / DATA SCIENCE TOOLS AND TECHNIQUES

TECHNICAL ANALYSIS USING R

Assignment 2 | Preliminary Stage

A reproducible stock-data import, statistics, and visualization workflow

STUDENT	**Mohammad Aljarad**
COURSE	CA-BDA400
ASSIGNMENT	A02 (5%)
REPOSITORY	https://github.com/maljarad30-coder/TechnicalAnalysis [VERIFY AFTER CREATION]

SUBMISSION CHECK / CURRENT STATUS

Prepared code and documentation framework. Execution evidence, actual outputs, and final repository link must be confirmed before submission.

1. Project scope and required deliverables

This preliminary stage creates a GitHub repository named TechnicalAnalysis and a separate Assignment2 folder. The workflow reads stock ticker symbols from a text file, downloads historical market observations, calculates descriptive statistics and a simple moving average, and displays results graphically.

- GitHub repository: TechnicalAnalysis, owned by the student.
- Stock symbol input: portfolio.txt with AAPL, MSFT and GOOGL on separate lines.
- Package installation script: 00_install_packages.R.
- Loading/statistics script: 01_load_and_statistics.R.
- Data display/plot script: 02_display_and_visualize.R.
- Technical report in Word, with screenshots; a PDF copy for the LMS submission.

2. Computational setup and R packages

R and RStudio have been used previously for this course. The assignment still requires documenting the installation/configuration as proof. For this project, install or verify quantmod, TTR, and zoo by running the setup script within RStudio.

3. Portfolio and data import

Each symbol is read from portfolio.txt, trimmed, normalized and validated. The script downloads approximately one year of daily price observations from Yahoo Finance using quantmod::getSymbols(). Each stock is saved as its own data frame with Date, Open, High, Low, Close and Volume. If an import fails, the script stops instead of reporting incomplete results as successful.

4. Statistical methods

Measure	Definition and interpretation
Mean close	Arithmetic average of daily closing prices.
Median close	Middle observed daily closing price.
Mode close	Most frequent closing price rounded to cents; blank if all values occur once.
Standard deviation	Sample standard deviation of daily closing prices.
20-day SMA	Average of the most recent 20 daily closing prices at each trading date.

5. Output specification

The second script prints the descriptive-statistics summary and the first/last six rows for every symbol. It creates outputs/statistics_summary.csv, one CSV with stock prices and SMA20 per symbol, and one PNG chart for each symbol. Actual market figures and plots must be generated in the student's RStudio session.

6. Reproduction instructions

1. Extract the project ZIP and open the Assignment2 directory in Windows File Explorer.

2. In RStudio use Session > Set Working Directory > Choose Directory and select Assignment2.

3. Run source("00_install_packages.R") and capture the package verification message.

4. Run source("02_display_and_visualize.R") and verify the final SUCCESS message.

5. Check all three symbols in the printed statistics table and inspect the outputs charts.

6. Upload the finished Assignment2 folder to GitHub and copy the live repository URL.

7. Add genuine screenshots below, save Word, then export the final submission PDF.

7. Screenshot evidence log

The screenshots below are required by the assignment rubric. Replace each placeholder with a screenshot from the student's actual computer or GitHub account, with readable text and relevant controls visible.

Evidence 01 | GitHub repository created

Show the actual TechnicalAnalysis repository name and student ownership.

EVIDENCE REQUIRED | Insert genuine screenshot here

Evidence 02 | RStudio working session

Show RStudio open with the Assignment2 working directory and an R version visible.

EVIDENCE REQUIRED | Insert genuine screenshot here

Evidence 03 | Package installation/verification

Show successful quantmod, TTR and zoo installation or verification in the RStudio Console.

EVIDENCE REQUIRED | Insert genuine screenshot here

Evidence 04 | Portfolio initialization

Show portfolio.txt opened, with AAPL, MSFT and GOOGL each on a separate line.

EVIDENCE REQUIRED | Insert genuine screenshot here

Evidence 05 | Successful stock import

Show Console data-download messages for all portfolio symbols.

EVIDENCE REQUIRED | Insert genuine screenshot here

Evidence 06 | Calculated statistics

Show the complete summary table containing mean, median, mode, standard deviation and 20-day SMA.

EVIDENCE REQUIRED | Insert genuine screenshot here

Evidence 07 | AAPL visualization

Show the generated AAPL closing-price and 20-day SMA chart.

EVIDENCE REQUIRED | Insert genuine screenshot here

Evidence 08 | MSFT and GOOGL visualizations

Show both remaining charts (use two screenshots if required for readability).

EVIDENCE REQUIRED | Insert genuine screenshot here

Evidence 09 | GitHub submitted files

Show Assignment2 with source files, screenshots, report and outputs uploaded to the repository.

EVIDENCE REQUIRED | Insert genuine screenshot here

8. Verification notes and limitations

All three stock symbols must load successfully and appear in the output. Internet access and the external Yahoo Finance service are required. The moving average is unavailable for the initial 19 observations. Historical stock values vary by data retrieval date. Any failed load or missing screenshot must be resolved before this report is submitted.

9. GitHub repository reference

Repository (expected after creation): https://github.com/maljarad30-coder/TechnicalAnalysis

FINAL VERIFICATION REQUIRED: Confirm this URL opens the correct repository, all required files are present, and outputs are genuine before submission.

10. References

- CDI College. BDA400 Assignment 2: Technical Analysis Using R, Preliminary Stage (instructor-provided PDF).
- GitHub Docs. Creating a repository for your project. docs.github.com.
- Posit Support. Working Directories and Workspaces in the RStudio IDE. support.posit.co.
- CRAN. quantmod and TTR package documentation. cran.r-project.org.